Example 8.15

Let $u(x,y)=e^{-2y}\cos(2x)$ for all $(x,y)\in\mathbb{R}^2$. Prove that u is a harmonic function in $\mathbb{R}^2$.

Solution

We need to show that u satisfies the Laplace's equation in $\mathbb{R}^2$. Observe that $u_x(x,y)=e^{-2y}(-2)\sin(2x)$ and hence $u_{xx}(x,y)=e^{-2y}(-2)(2)\cos(2x)$.

Similarly, $u_y(x,y)=e^{-2y}(-2)\cos(2x)$ and $u_{yy}(x,y)=(-2)(-2)e^{-2y}\cos(2x)$.

Thus, $u_{xx}+u_{yy}=-4e^{-2y}\cos(2x)+4e^{-2y}\cos(2x)=0$.

EXERCISE 8.4

1. Find the partial derivatives of the following functions at the indicated points.

   (i) $f(x,y)=3x^2-2xy+y^2+5x+2,\quad (2,-5)$

   (ii) $g(x,y)=3x^2+y^2+5x+2,\quad (1,-2)$

   (iii) $h(x,y,z)=x\sin(xy)+z^2x,\quad \left(2,\frac{\pi}{4},1\right)$

   (iv) $G(x,y)=e^{x+3y}\log(x^2+y^2),(-1,1)$

2. For each of the following functions find the f_x, f_y, and show that $f_{xy}=f_{yx}$.

   (i) $f(x,y)=\dfrac{3x}{y+\sin x}$ (ii) $f(x,y)=\tan^{-1}\left(\dfrac{x}{y}\right)$ (iii) $f(x,y)=\cos(x^2-3xy)$

3. If $U(x,y,z)=\dfrac{x^2+y^2}{xy}+3z^2y$, find $\dfrac{\partial U}{\partial x},\dfrac{\partial U}{\partial y}$, and $\dfrac{\partial U}{\partial z}$.

4. If $U(x,y,z)=\log(x^3+y^3+z^3)$, find $\dfrac{\partial U}{\partial x}+\dfrac{\partial U}{\partial y}+\dfrac{\partial U}{\partial z}$.

5. For each of the following functions find the g_{xy}, g_{xx}, g_{yy} and g_{yx}.

   (i) $g(x,y)=xe^y+3x^2y$ (ii) $g(x,y)=\log(5x+3y)$

   (iii) $g(x,y)=x^2+3xy-7y+\cos(5x)$

6. Let $w(x,y,z)=\dfrac{1}{\sqrt{x^2+y^2+z^2}},(x,y,z)\neq(0,0,0)$. Show that $\dfrac{\partial^2 w}{\partial x^2}+\dfrac{\partial^2 w}{\partial y^2}+\dfrac{\partial^2 w}{\partial z^2}=0$.

7. If $V(x,y)=e^x(x\cos y-y\sin y)$, then prove that $\dfrac{\partial^2 V}{\partial x^2}+\dfrac{\partial^2 V}{\partial y^2}=0$.

8. If $w(x,y)=xy+\sin(xy)$, then prove that $\dfrac{\partial^2 w}{\partial y\partial x}=\dfrac{\partial^2 w}{\partial x\partial y}$.

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